

Aeroelastic stability of a 2-DOF mechanical system.

1 Introduction

Aeroelasticity is concerned with the effect produced on mechanical systems by their interaction with an air flow. We consider the case of a rigid body subjected to the forces produced by air flowing around it: we will show below that the forces produced by the air flow on the body depend on the state of the system, i.e. on the position and velocity of the body. Like in the case of other state-dependent forces, the effect of aerodynamic forces on the moving body can be seen, for a small motion around a given equilibrium position, as corresponding to additional stiffness and damping terms, hence the name of “aeroelasticity”, i.e. the presence of additional stiffness (“elastic” according to a somehow improper terminology) and damping terms generated by the air flow.

In this lesson, we first introduce a mathematical model for the forces caused on the moving body by the air flow, according to the quasi-static theory (note, this is just one out of several models that can be used in aeroelasticity problems) and then we use this model to analyse the stability of a 2-dof system.

2 Model of aerodynamic forces

We consider first the case of a rigid body which is kept at a steady position and subjected to an air flow, see Figure 2.1. For the sake of simplicity, we consider the shape of the body to be symmetric with respect to an axis of symmetry whose direction is “nearly aligned” to the direction of the vector representing the velocity of air particles. The air flow generates on the body a distribution of normal pressure and tangential tractions. Since we are considering the body as rigid, we just care about the net force (i.e. the vector sum of the distributed forces acting on the body) and the total moment of the distributed aerodynamic forces, which are shown in the lower part of the figure. It is customary to describe the net vector applied by the flow to the body in terms of a component alongside the air flow (parallel to the vector $\vec{V}$ defining the direction of the air particles) and a component perpendicular to $\vec{V}$. These two components are called, respectively, the *drag force* F_D and *lift force* F_L .

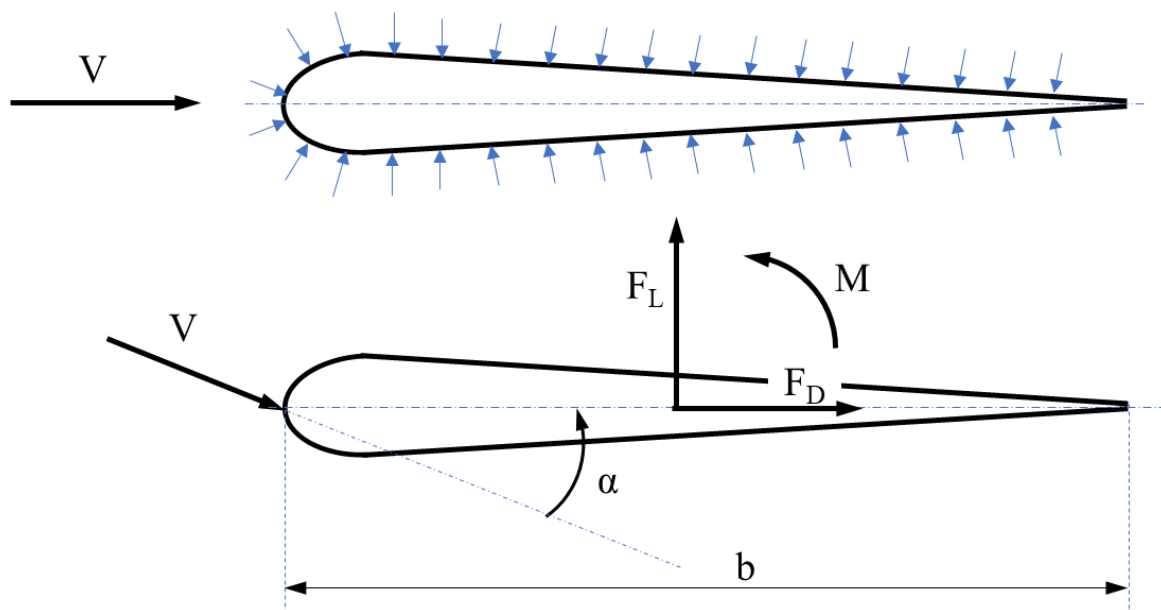


Figure 2.1: Forces produced on a rigid body by an incoming air flow.

The drag and lift force components and the aerodynamic moment M can be defined according to the following expressions, valid when the rigid body is still:

$$\begin{aligned} F_D &= \frac{1}{2} \rho V^2 S c_D(\alpha) \\ F_L &= \frac{1}{2} \rho V^2 S c_L(\alpha) \\ M &= \frac{1}{2} \rho V^2 S b c_M(\alpha) \end{aligned} \quad (2.1)$$

where ρ is air density, V is the speed of the air flow, S is a reference surface in the rigid body, b is the chord of the body (see Figure 2.1) and finally c_D , c_L and c_M are three non-dimensional aerodynamic coefficients depending on the shape of the body and on the angle α formed by the direction of the air flow and a characteristic direction in the body (in the case of a symmetric profile, this latter direction is the symmetry axis). Angle α is called the *angle of attack* of the flow.

The curves describing how the aerodynamic coefficients c_D , c_L and c_M vary with the angle of attack can be obtained from measurements performed in a wind tunnel or can be obtained by solving numerically a system of partial derivative differential equations describing the motion of the air flow around the body. For a symmetric profile like the one shown in Figure 2.1, the qualitative shape of the curves is the one shown in Figure 2.2. Note the curves in this figure are for the “nose down” sign convention for the attack angle, i.e. a positive value of angle α corresponds to a rotation of the body where the leading edge of the profile moves down with respect to the incoming flow.

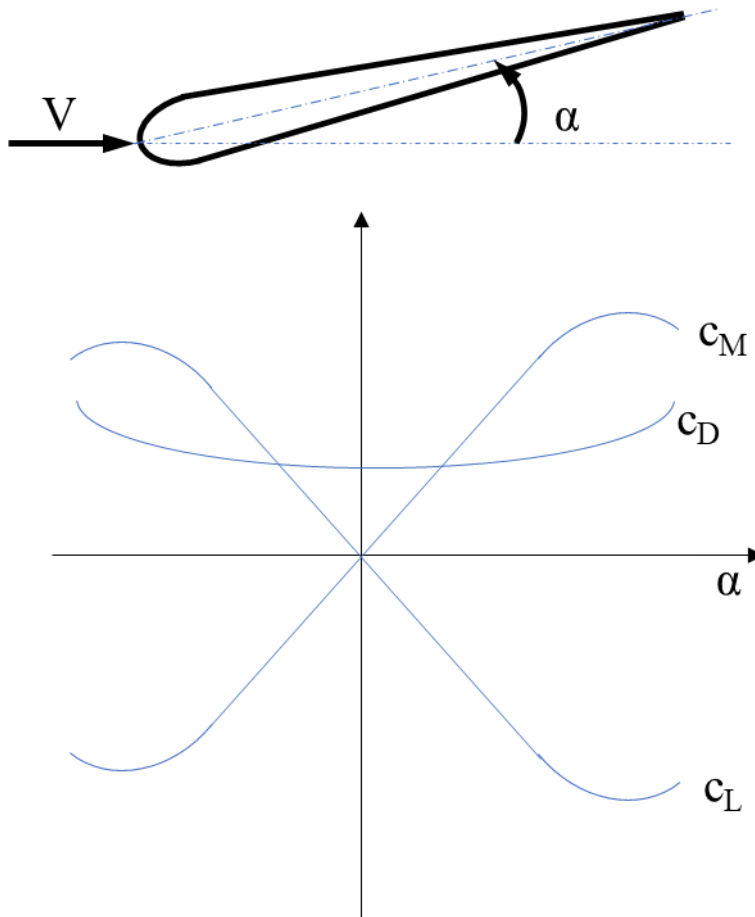


Figure 2.2: Aerodynamic coefficients for a symmetric airfoil.

Equation (2.1) is known under the name of “static theory”, as it refers to the case of the body standing still while being hit by the air flow. In case the body is moving instead, the aerodynamic forces generated on the body depend both on the air flow and on the motion of the body. One possible way to describe these forces is according to the “quasi-static theory”, i.e. an extension of the static theory

to a case when the rigid body performs a motion which is sufficiently “slow”. In the quasi-static theory, the expression of the aerodynamic forces and moments according to the static theory is modified replacing the absolute wind speed vector by a relative speed vector, so that the effect of both the absolute motion of air particles and of the body can be considered. The quasi-static theory can be considered valid as long as the motion of the body in the flow is sufficiently slow. More precisely, this means the time corresponding to one oscillation in the motion of the body shall be much longer than the time required to a particle to run through the body. This condition can be turned to quantitative according to the following equation:

$$f_r = \frac{1}{f} \frac{b}{V} \ll 1 \quad (2.2)$$

with f_r the *reduced frequency*, a non-dimensional parameter defining the ratio of the two characteristic times mentioned above. We assume below the condition (2.2) is satisfied.

The vector sum defining the relative wind velocity vector $\vec{V}_R$ according to the quasi-static theory is shown in Figure 2.3: assuming the rigid body motion is described by coordinate x and θ as shown in the figure, a different relative speed $\vec{V}_R$ is obtained at any point in the surface of the body. To solve this issue, the quasi-static theory considers the relative speed vector for a reference point which is typically located close to the leading edge of the profile, whose position is defined through the distance b_1 of the point from the body's centre of mass.

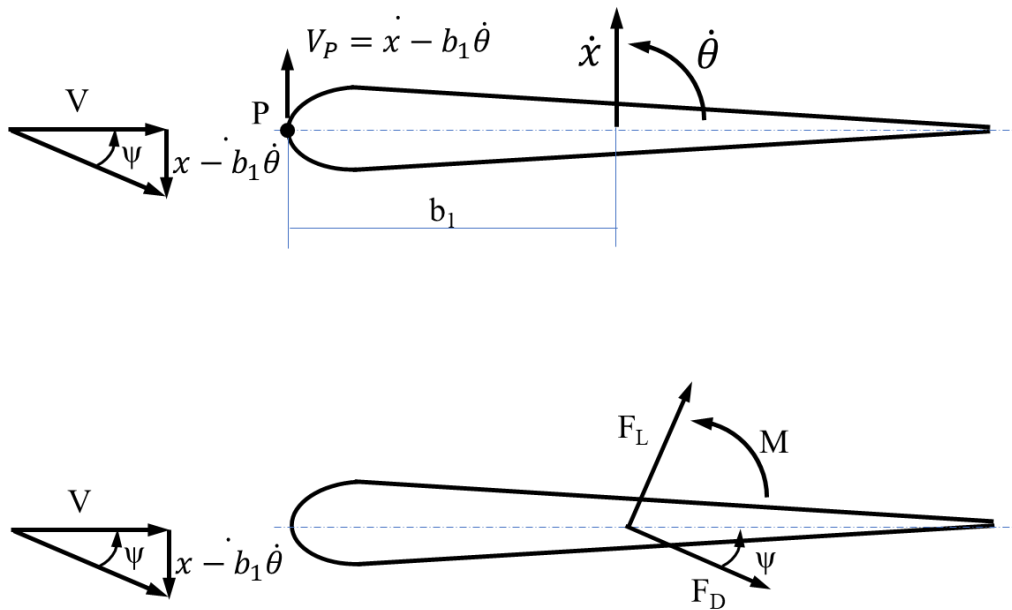


Figure 2.3: Velocity of air relative to the moving profile and direction of drag and lift forces according to the quasi-static theory.

Hence, according to the quasi-static theory, the aerodynamic forces are defined as:

$$\begin{aligned} F_D &= \frac{1}{2} \rho V_R^2 S c_D(\alpha) \\ F_L &= \frac{1}{2} \rho V_R^2 S c_L(\alpha) \\ M &= \frac{1}{2} \rho V_R^2 S b c_M(\alpha) \end{aligned} \quad (2.3)$$

with:

$$\begin{aligned}
 V_R^2 &= V^2 + (\dot{x} - b_1 \dot{\theta})^2 \\
 \alpha &= \theta + \psi \\
 \psi &= \tan^{-1} \left(\frac{\dot{x} - b_1 \dot{\theta}}{V} \right)
 \end{aligned}
 \tag{2.4}$$

We note from Figure 2.3 that considering the relative velocity vector in place of the absolute air velocity vector means the directions of the drag and lift components in the aerodynamic forces are varying with the motion of the body and are defined by the angle ψ defined by Eq. (2.4).

3 Stability of a 2-dof airfoil subject to air flow.

In this section, we analyse the motion of a 2-dof rigid body (coordinates x and θ as already defined in Figure 2.3) hit by an air flow whose absolute velocity is constant in magnitude and direction. The body is supported by springs and dampers as shown in Figure 3.1. Figure 3.2 shows the forces acting on the rigid body, under the assumption of small motion of the body. The equations of motion for the 2-dof system are:

$$\begin{aligned}
 m\ddot{x} + c\dot{x} + kx &= -F_D \sin \psi + F_L \cos \psi \\
 J\ddot{\theta} + cd^2\dot{\theta} + kd^2\theta &= M
 \end{aligned}
 \tag{3.1}$$

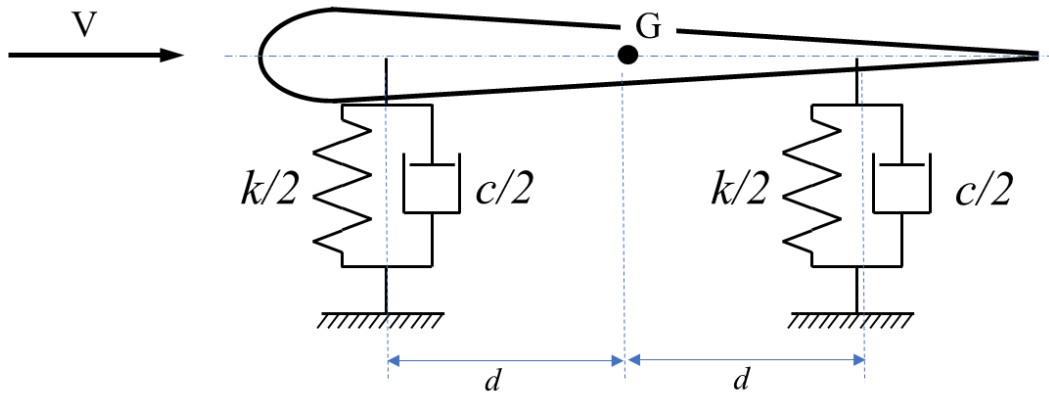


Figure 3.1: rigid wing with 2 dofs subject to air flow.

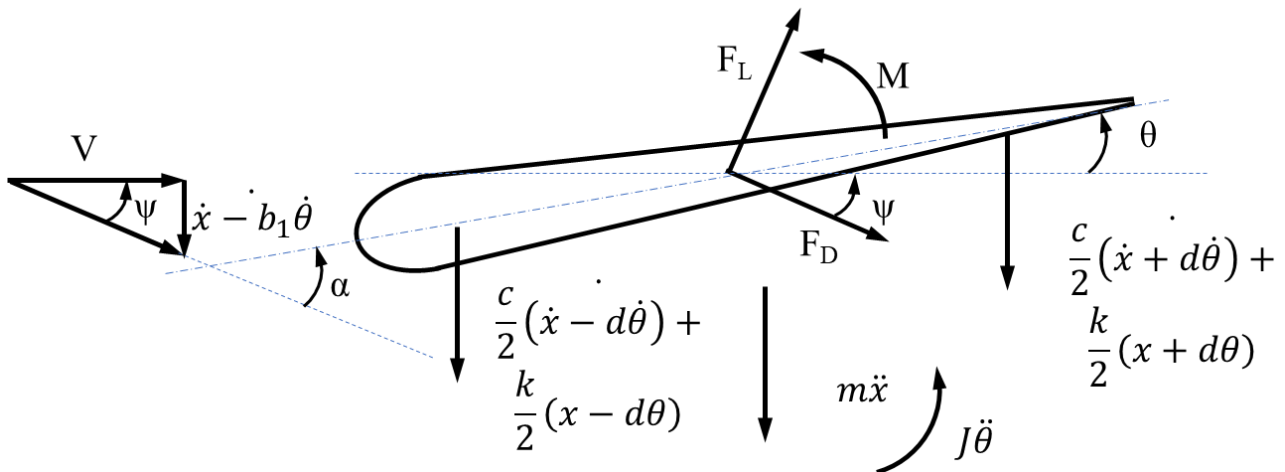


Figure 3.2: forces acting on the 2 DOFs system.

First of all, we analyse the equilibrium position(s) of the 2-dof system. We note that, according to Eq. (2.4), when the body is at an equilibrium position $\psi=0$ and $\alpha=0$. Hence, an equilibrium position of the system is defined according to:

$$\begin{aligned} kx_e &= F_L = \frac{1}{2} \rho V^2 S c_L(\mathcal{G}_e) \\ kd^2\theta_e &= M = \frac{1}{2} \rho V^2 S b c_M(\mathcal{G}_e) \end{aligned} \quad (3.2)$$

For a symmetric profile having aerodynamic coefficients as shown in Figure 2.2 we see that the equilibrium condition corresponding to Eq. (3.2) is satisfied by:

$$\begin{aligned} \mathcal{G}_e &= 0 \\ x_e &= 0 \end{aligned} \quad (3.3)$$

Note it is possible that two other equilibrium positions are obtained for a large positive or negative rotation $\mathcal{G}_e$ of the profile, however we are mostly interested here in studying the equilibrium position corresponding to zero rotation of the profile.

Based on Lyapounov's theorem, we linearise the equations of motion of the system in the neighbourhood of the equilibrium position of the system defined by Eq. (3.3). The linearised expressions corresponding to the non-linear expressions in Eq. (2.3) are:

$$\begin{aligned} V_R^2 &= V^2 + (\dot{x} - b_1 \dot{\theta})^2 \cong V^2 \\ \psi &= \tan^{-1} \left(\frac{\dot{x} - b_1 \dot{\theta}}{V} \right) \cong \frac{\dot{x} - b_1 \dot{\theta}}{V} \\ \alpha &= \theta + \psi \cong \theta + \frac{\dot{x} - b_1 \dot{\theta}}{V} \end{aligned} \quad (3.4)$$

Furthermore, noting that at the equilibrium position we have $\alpha=0$, we approximate the functions $c_D = c_D(\alpha)$, $c_L = c_L(\alpha)$, $c_M = c_M(\alpha)$ to their power series expansion truncated to first order terms in the neighbourhood of $\alpha=0$:

$$\begin{aligned} c_D(\alpha) &\cong c_D(0) + \left. \frac{dc_D}{d\alpha} \right|_{\alpha=0} \alpha = C_{D0} \\ c_L(\alpha) &\cong c_L(0) + \left. \frac{dc_L}{d\alpha} \right|_{\alpha=0} \alpha = K_{L0} \alpha \\ c_M(\alpha) &\cong c_M(0) + \left. \frac{dc_M}{d\alpha} \right|_{\alpha=0} \alpha = K_{M0} \alpha \end{aligned} \quad (3.5)$$

where we note that for a symmetric profile the lift and moment coefficients and the derivative of the drag coefficient are all zero at $\alpha=0$ and we introduce the following notation:

$$\begin{aligned} C_{D0} &= c_D(0) \\ K_{L0} &= \left. \frac{dc_L}{d\alpha} \right|_{\alpha=0} \\ K_{M0} &= \left. \frac{dc_M}{d\alpha} \right|_{\alpha=0} \end{aligned}$$

Using Eq.s (3.4) and (3.5) we get the linearised expressions for the aerodynamic forces and moments:

$$\begin{aligned}
F_D &= \frac{1}{2} \rho V^2 S C_{D0} \\
F_L &= \frac{1}{2} \rho V^2 S K_{L0} \alpha = \frac{1}{2} \rho V^2 S K_{L0} \left(\theta + \frac{\dot{x} - b_1 \dot{\theta}}{V} \right) \\
M &= \frac{1}{2} \rho V_R^2 S b K_{M0} \alpha = \frac{1}{2} \rho V^2 S b K_{M0} \left(\theta + \frac{\dot{x} - b_1 \dot{\theta}}{V} \right)
\end{aligned} \tag{3.6}$$

Next, we bring Eq. (3.6) in Eq. (3.1), noting that for small values of angle ψ we can use the approximations $\cos(\psi) \cong 1$ and $\sin(\psi) \cong \psi$:

$$\begin{aligned}
m\ddot{x} + c\dot{x} + kx &= -\frac{1}{2} \rho V^2 S C_{D0} \psi + \frac{1}{2} \rho V^2 S K_{L0} \alpha = -\frac{1}{2} \rho V^2 S C_{D0} \left(\frac{\dot{x} - b_1 \dot{\theta}}{V} \right) + \frac{1}{2} \rho V^2 S K_{L0} \left(\theta + \frac{\dot{x} - b_1 \dot{\theta}}{V} \right) \\
J\ddot{\theta} + cd^2\dot{\theta} + kd^2\theta &= \frac{1}{2} \rho V^2 S b K_{M0} \alpha = \frac{1}{2} \rho V^2 S b K_{M0} \left(\theta + \frac{\dot{x} - b_1 \dot{\theta}}{V} \right)
\end{aligned} \tag{3.7}$$

Finally, we re-arrange Eq. 3.7 in matrix form as:

$$[m]\ddot{\underline{x}} + ([c_S] + [c_F])\dot{\underline{x}} + ([k_S] + [k_F])\underline{x} = \underline{0} \tag{3.7}$$

with:

$$\begin{aligned}
\underline{x} &= \begin{Bmatrix} x \\ \theta \end{Bmatrix} \\
[m] &= \begin{bmatrix} m & 0 \\ 0 & J \end{bmatrix} \\
[c_S] &= \begin{bmatrix} c & 0 \\ 0 & cd^2 \end{bmatrix} \\
[k_S] &= \begin{bmatrix} k & 0 \\ 0 & kd^2 \end{bmatrix} \\
[c_F] &= \frac{1}{2} \rho V S \begin{bmatrix} C_{D0} - K_{L0} & -b_1(C_{D0} - K_{L0}) \\ -bK_{M0} & bb_1K_{M0} \end{bmatrix} \\
[k_F] &= \frac{1}{2} \rho V^2 S \begin{bmatrix} 0 & -K_{L0} \\ 0 & -bK_{M0} \end{bmatrix}
\end{aligned} \tag{3.8}$$

We note that the $[c_F]$ and $[k_F]$ matrices are both non-symmetric. Furthermore, matrix $[k_F]$ is non positive definite. Hence, different types of instability may take place:

- 1) **1-dof vertical**: a dynamic instability corresponding to an **expansive vertical oscillation** of the profile is possible in case $C_{D0} - K_{L0} < 0$. This is not the case for the symmetric profile whose aerodynamic coefficient are shown in Figure 2.2, because we see in that case K_{L0} is negative. However, this type of instability may **take place for a “bluff” body**, i.e. a body having a large frontal surface exposed to the wind. For this type of profile a positive value of K_{L0} can be obtained so that $C_{D0} - K_{L0} < 0$.
- 2) **1-dof torsional**: a dynamic instability may also take place in the torsional motion of the profile, in case $K_{M0} < 0$. Again, as shown by Figure 2.2, this is not the case for an elongated profile like the one considered here, but a negative value of K_{M0} may be found in some cases for bluff bodies.
- 3) **1-dof torsional**: considering the aerodynamic coefficients in Figure 2.2, we note K_{M0} is positive and hence the diagonal term in second row of matrix $[k_F]$ is negative. This means

that, when wind speed is sufficiently high, the total stiffness related to the rotation θ of the profile is negative. In this case, a divergence-type instability takes place, involving the rotation of the body. The value of wind speed required to cause this type of instability is in general very large, so it is quite unlikely that this instability takes place.

- 4) 2-dof instability: despite the ‘classical condition’ for flutter instability is not met, note that $k_{Txy}k_{Tyx} = 0$, we see that the effect of aeroelastic forces is to lower the natural frequency associated with the torsional oscillation of the profile. Quite often, for zero wind speed, i.e. owing just to the structural terms in the mass and stiffness matrices, the torsional natural frequency is higher than the one associated to the vertical oscillation of the profile. Hence, we expect that at some non-zero value of wind speed (lower than the one required to cause the divergence-type instability mentioned at point 3 above) the two natural frequencies related to the vertical and torsional oscillations of the body synchronise. In this situation, a synchronisation also takes place between the net force in vertical direction acting on the profile (which is mainly caused by the rotation of the profile) and the vertical oscillation of the body. In this condition, the in-phase vertical force and vertical displacement lead to a positive work performed by the aerodynamic forces and hence to continuous introduction of energy in the oscillation of the system that becomes therefore expansive. This case is called “flutter” instability, despite it is slightly different from the case of the railway wheelset.